

Worksheet — Relations and Functions-I (NIOs Official)

Total marks: 30 · Questions: 10

Instructions: This is the NIOS Senior Secondary Mathematics (311) official Worksheet-2 for Lesson 2: Relations and Functions-I. Attempt all questions. Show every step in roster form and clearly state the domain, range and co-domain where asked.

Q1. Write a relation R 'is capital of' from set A to set B , where A has five capitals and B has five states of India. [2 marks]

Q2. Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. Find $A \times B$ and $B \times A$ and verify $A \times B \neq B \times A$. [3 marks]

Q3. Let R be a relation from N to N defined by $R = \{(x, y) : x + 2y = 10, x, y \in N\}$. Find (i) R in roster form, (ii) domain and range of R . [3 marks]

Q4. For the function $f(x) = 3x - 2$, find the range when the domain is $\{-2, 1, 2, 3\}$. [2 marks]

Q5. Find the domain of the functions (i) $y = 1/(x^2 - 5x + 7)$ for $x \in R$, and (ii) $y = 1/(x - 2)(x - 3)$ for $x \in R$. [3 marks]

Q6. Let $A = \{1, 2, 3, 4, 5\}$ and R be the relation on A defined by $\{(a, b) : a, b \in A \text{ and } a \text{ divides } b\}$.
(i) Write R in roster form. (ii) Find domain and range of R . [3 marks]

Q7. Let $f(x) = 3x - 2$ and $g(x) = 2x + 1$ be two real functions. Find (i) $(f + g)(x)$, (ii) $(f - g)(x)$. [3 marks]

Q8. Given $A = \{1, 2, 3\}$, $B = \{3, 4\}$ and $C = \{4, 5\}$, verify (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$ and (ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$. [4 marks]

Q9. If $f(x) = x + 5$, $x \in \{0, 1, 2, 3, \dots, 6\}$, find the range of the function and represent it via an arrow diagram. [3 marks]

Q10. Cite suitable examples of odd functions and even functions. Draw (in words) the graph of any one odd and any one even function, and write your observations. [4 marks]

Answer Key

Worksheet — Relations and Functions-I (NIOS Official)

Q1. Write a relation R 'is capital of' from set A to set B, where A has five capitals and B has five states of India.

Answer: Take $A = \{\text{Mumbai, Bengaluru, Chennai, Lucknow, Bhopal}\}$ and $B = \{\text{Maharashtra, Karnataka, Tamil Nadu, Uttar Pradesh, Madhya Pradesh}\}$. Then $R = \{(\text{Mumbai, Maharashtra}), (\text{Bengaluru, Karnataka}), (\text{Chennai, Tamil Nadu}), (\text{Lucknow, Uttar Pradesh}), (\text{Bhopal, Madhya Pradesh})\}$. R is the relation 'is capital of'. Domain = A, range = B in this case.

Any five (capital, state) pairs where the first really is the capital of the second are acceptable.

SHORT_ANSWER · EASY · 12-NIOS-MAT-RF-02

Q2. Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. Find $A \times B$ and $B \times A$ and verify $A \times B \neq B \times A$.

Answer: $A \times B = \{(1,a), (1,b), (2,a), (2,b), (3,a), (3,b)\}$ (6 ordered pairs). $B \times A = \{(a,1), (a,2), (a,3), (b,1), (b,2), (b,3)\}$ (6 ordered pairs). Both have $6 = 3 \times 2$ elements (consistent with $n(A \times B) = n(A) \cdot n(B)$) but the ELEMENTS differ — e.g. $(1,a) \in A \times B$ but $(1,a) \notin B \times A$. So $A \times B \neq B \times A$ (they're not the same set of ordered pairs).

Cartesian product is not commutative because ordered pairs care about which element comes first.

SHORT_ANSWER · MEDIUM · 12-NIOS-MAT-RF-01

Q3. Let R be a relation from N to N defined by $R = \{(x, y) : x + 2y = 10, x, y \in \mathbb{N}\}$. Find (i) R in roster form, (ii) domain and range of R.

Answer: Solve $x + 2y = 10$ with $x, y \in \mathbb{N}$ (positive integers): $y = (10 - x)/2$, requires $10 - x$ even and positive $\Rightarrow x \in \{2, 4, 6, 8\}$ giving $y \in \{4, 3, 2, 1\}$. (i) $R = \{(2, 4), (4, 3), (6, 2), (8, 1)\}$. (ii) Domain = $\{2, 4, 6, 8\}$. Range = $\{1, 2, 3, 4\}$.

Generate ordered pairs satisfying $x + 2y = 10$ with both x, y natural numbers.

SHORT_ANSWER · MEDIUM · 12-NIOS-MAT-RF-03

Q4. For the function $f(x) = 3x - 2$, find the range when the domain is $\{-2, 1, 2, 3\}$.

Answer: $f(-2) = 3(-2) - 2 = -8$. $f(1) = 3 - 2 = 1$. $f(2) = 6 - 2 = 4$. $f(3) = 9 - 2 = 7$. Range = $\{-8, 1, 4, 7\}$.

Substitute each domain value into f and collect the outputs.

SHORT_ANSWER · EASY · 12-NIOS-MAT-RF-04

Q5. Find the domain of the functions (i) $y = 1/(x^2 - 5x + 7)$ for $x \in \mathbb{R}$, and (ii) $y = 1/(x - 2)(x - 3)$ for $x \in \mathbb{R}$.

Answer: (i) Need $x^2 - 5x + 7 \neq 0$. Discriminant = $25 - 28 = -3 < 0 \Rightarrow$ no real roots. So $x^2 - 5x + 7 > 0$ for ALL real x. Domain = $\mathbb{R}$. (ii) Need $(x - 2)(x - 3) \neq 0 \Rightarrow x \neq 2$ and $x \neq 3$. Domain = $\mathbb{R} - \{2, 3\} = \{x : x \in \mathbb{R} \text{ and } x \neq 2, x \neq 3\}$.

A real function's domain is the set of x-values for which the formula is defined. For $1/g(x)$ we exclude zeros of g.

SHORT_ANSWER · MEDIUM · 12-NIOS-MAT-RF-08

Q6. Let $A = \{1, 2, 3, 4, 5\}$ and R be the relation on A defined by $\{(a, b) : a, b \in A \text{ and } a \text{ divides } b\}$.
 (i) Write R in roster form. (ii) Find domain and range of R .

Answer: List all pairs with $a \mid b$ (and $a, b \in A$): $a=1$: $1 \mid 1, 1 \mid 2, 1 \mid 3, 1 \mid 4, 1 \mid 5 \rightarrow (1,1), (1,2), (1,3), (1,4), (1,5)$
 $a=2$: $2 \mid 2, 2 \mid 4 \rightarrow (2,2), (2,4)$ $a=3$: $3 \mid 3 \rightarrow (3,3)$ $a=4$: $4 \mid 4 \rightarrow (4,4)$ $a=5$: $5 \mid 5 \rightarrow (5,5)$ $R = \{(1,1), (1,2), (1,3), (1,4), (1,5), (2,2), (2,4), (3,3), (4,4), (5,5)\}$. Domain = $\{1, 2, 3, 4, 5\}$ (every a appears as a first component). Range = $\{1, 2, 3, 4, 5\}$ (every b appears as a second component).

Generate pairs where 'a divides b' ($a \mid b$ means b/a is a positive integer).

SHORT_ANSWER · MEDIUM · 12-NIOS-MAT-RF-03

Q7. Let $f(x) = 3x - 2$ and $g(x) = 2x + 1$ be two real functions. Find (i) $(f + g)(x)$, (ii) $(f - g)(x)$.

Answer: (i) $(f + g)(x) = f(x) + g(x) = (3x - 2) + (2x + 1) = 5x - 1$. (ii) $(f - g)(x) = f(x) - g(x) = (3x - 2) - (2x + 1) = x - 3$. Both are defined for all $x \in \mathbb{R}$ (domain = $\mathbb{R}$), since f and g themselves are defined everywhere.

Pointwise addition / subtraction: combine the formulas algebraically.

SHORT_ANSWER · MEDIUM · 12-NIOS-MAT-RF-07

Q8. Given $A = \{1, 2, 3\}$, $B = \{3, 4\}$ and $C = \{4, 5\}$, verify (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$ and (ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

Answer: (i) $B \cup C = \{3, 4, 5\}$; $A \times (B \cup C) = \{(1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5)\}$ (9 pairs). $A \times B = \{(1,3), (1,4), (2,3), (2,4), (3,3), (3,4)\}$; $A \times C = \{(1,4), (1,5), (2,4), (2,5), (3,4), (3,5)\}$; $(A \times B) \cup (A \times C) = \{(1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5)\}$ — same 9 pairs ✓. (ii) $B \cap C = \{4\}$; $A \times (B \cap C) = \{(1,4), (2,4), (3,4)\}$. $(A \times B) \cap (A \times C)$: common pairs are exactly those with second component 4 = $\{(1,4), (2,4), (3,4)\}$ ✓. Both distributive laws verified.

These are the distributive laws of Cartesian product over union/intersection.

SHORT_ANSWER · MEDIUM · 12-NIOS-MAT-RF-01

Q9. If $f(x) = x + 5$, $x \in \{0, 1, 2, 3, \dots, 6\}$, find the range of the function and represent it via an arrow diagram.

Answer: $f(0) = 5$; $f(1) = 6$; $f(2) = 7$; $f(3) = 8$; $f(4) = 9$; $f(5) = 10$; $f(6) = 11$. Range = $\{5, 6, 7, 8, 9, 10, 11\}$. Arrow diagram (described in words): on the left, label seven dots 0, 1, 2, 3, 4, 5, 6 (the domain). On the right, label seven dots 5, 6, 7, 8, 9, 10, 11 (the range). Draw arrows from $0 \rightarrow 5$, $1 \rightarrow 6$, $2 \rightarrow 7$, $3 \rightarrow 8$, $4 \rightarrow 9$, $5 \rightarrow 10$, $6 \rightarrow 11$. Each domain element has exactly one outgoing arrow, confirming f is a well-defined function.

Substitute every domain value, list the outputs, then describe the arrow picture connecting input to output.

SHORT_ANSWER · EASY · 12-NIOS-MAT-RF-04

Q10. Cite suitable examples of odd functions and even functions. Draw (in words) the graph of any one odd and any one even function, and write your observations.

Answer: **EVEN functions** ($f(-x) = f(x)$): $f(x) = x^2$; $f(x) = |x|$; $f(x) = \cos x$; $f(x) = x^4 + 3$. **ODD functions** ($f(-x) = -f(x)$): $f(x) = x$; $f(x) = x^3$; $f(x) = \sin x$; $f(x) = x^3 + 2x$. Graph of an **EVEN** function (e.g. $f(x) = x^2$): a parabola opening upward, symmetric about the y-axis. For any x , the point (x, x^2) and $(-x, x^2)$ lie at the same height — the graph is its own mirror image across the y-axis. Graph of an **ODD** function (e.g. $f(x) = x^3$): an S-shaped curve passing through the origin. For any x , the point (x, x^3) and $(-x, -x^3)$ are diagonally opposite — rotate the curve by 180° about the origin and it lies on top of itself. **Observations:** (a) Every even function has y-axis symmetry; every odd function has origin symmetry. (b) Odd functions necessarily satisfy $f(0) = 0$ (because $f(-0) = -f(0) \Rightarrow f(0) = 0$). (c) Sum of two evens is even; sum of two odds is odd; sum of an even and an odd is generally neither. (d) **Product:** even $\times$ even = even; odd $\times$ odd = even; even $\times$ odd = odd.

Provide examples, describe symmetry of graphs, and note algebraic properties.

LONG_ANSWER · MEDIUM · 12-NIOS-MAT-RF-06